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Correct
Answer

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Practice Exam - 1

Question 107 of 117



A 47-year-old man with fulminant myocarditis is supported with femoral-femoral VA-ECMO and epinephrine 6 mcg/min. Echocardiography shows the aortic valve no longer opens and there is spontaneous echo contrast ("smoke") in the left ventricle. Over 24 hours, chest radiograph shows worsening bilateral pulmonary edema, and pulmonary artery diastolic pressure rises from 18 to 30 mmHg despite unchanged ECMO flow.

What is the most appropriate next step in management?

- A. Increase VA-ECMO circuit flow
- ☒ B. Place a percutaneous LV venting device (e.g., Impella) in addition to VA-ECMO ("ECPELLA")
- C. Initiate norepinephrine dose to try to allow for AV valve opening
- D. Aggressive intravenous diuresis alone
- E. Decrease ECMO flow without additional venting

Commentary

References

Content Code (Blueprint): Critical Care Cardiology — Mechanical Circulatory Support Complications (AHFTC Blueprint)

Task: Management

Teaching Point: Peripheral VA-ECMO increases left ventricular afterload by delivering retrograde aortic flow; in the setting of severe LV dysfunction this can cause LV distention, elevated left atrial/pulmonary pressures, pulmonary edema, and intracardiac stasis with thrombus risk, requiring an LV venting strategy.

Correct Answer (Best Answer Choice): Place a percutaneous LV venting device (e.g., Impella) in addition to VA-ECMO ("ECPELLA")

Distractors (Plausible Incorrect Options):

- Increase VA-ECMO circuit flow
- Increase norepinephrine dose to raise mean arterial pressure
- Aggressive intravenous diuresis alone
- Decrease ECMO flow without additional venting

Rationale:

Correct answer B: (Impella/ECPELLA or LV vent): A non-ejecting, distended left ventricle on VA-ECMO causes pulmonary congestion and risks intracardiac thrombus and pulmonary hemorrhage. Active unloading with a transvalvular axial-flow pump, surgical vent, or atrial septostomy decompresses the LV and restores forward flow across the aortic valve. Further, LV unloading improves the chance for myocardial recovery which is the goal in this patient with de novo acute heart failure.

Incorrect answers:

- A. Increasing ECMO flow: This further raises retrograde aortic pressure and LV afterload, worsening distention rather than relieving it.
- B. Correct answers
- C. Starting norepinephrine can further raise MAP and the resistance the native LV must eject against, worsening afterload mismatch and distention. Further, the patient is not ejecting despite high dose epinephrine.
- D. Diuresis alone: While volume removal may help, it does not address the mechanical afterload mismatch causing a non-ejecting, distended ventricle, and the patient will likely not be diuretic-responsive in this state.
- E. Decreasing ECMO flow without venting: Reducing flow may lower afterload but sacrifices systemic perfusion support and does not reliably resolve distention if the LV remains unable to eject.

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